

PDE'S

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1. 22/4/09: Dirichlet Problem

Comments about solving the Dirichlet energy minimisation problem using Sobolev spaces.

In particular, note that *the full minimising sequence converges (not just a subsequence), and the convergence is in the (strong) $W^{1,2}$ sense.*

1.1. Sobolev Spaces. Suppose $\Omega \subset \mathbb{R}^n$ where Ω is a bounded domain (i.e. open and connected).

Definition 1.1.

$$\begin{aligned} W^{1,2}(\Omega) &= \{u \in L^2(\Omega) : Du \in L^2(\Omega), \text{ where } Du \text{ is the weak derivative}\}, \\ W_0^{1,2}(\Omega) &= \text{closure of } C_0^1(\Omega) \text{ in } W^{1,2}(\Omega). \end{aligned}$$

See [GT83, §§7.3, 7.5]

By “ Du is the weak derivative” we mean for $i = 1, \dots, n$ there exists $v_i \in L^2(\Omega)$ s.t.

$$(1) \quad \int_{\Omega} u D_i \phi = - \int_{\Omega} v_i \phi \quad \forall \phi \in C_0^1(\Omega),$$

in which case v_i is unique and is denoted by $D_i u$.

Remark 1.2 (An expression for the $W^{1,2}(\Omega)$ norm). From (1), if $u \in L^2(\Omega)$ then $D_i u$ exists and is in $L^2(\Omega)$, i.e. $u \in W^{1,2}(\Omega)$, iff the right side of (2) is finite, in which case

$$(2) \quad \int_{\Omega} |D_i u|^2 = \sup \left\{ \int_{\Omega} u D_i \phi : \phi \in C_0^1(\Omega), \|\phi\|_{L^2} \leq 1 \right\}.$$

Remark 1.3 ($W^{1,2}$ Inner Product). Note that $W_0^{1,2}(\Omega)$ is an inner product space with

$$(f, g)_{W^{1,2}} = \int_{\Omega} Df \cdot Dg.$$

Similarly for $W^{1,2}(\Omega)$ if we take equivalence classes modulo constant functions.

Remark 1.4 (Weak Convergence). One says $u_n \rightharpoonup u$ (weakly) in $L^2(\Omega)$ if $\int_{\Omega} u_n \phi \rightarrow \int_{\Omega} u \phi$ for all ϕ in some dense subset of $L^2(\Omega)$. It follows

$$\int_{\Omega} |u|^2 \leq \liminf \int_{\Omega} |u_n|^2.$$

One says $u_n \rightharpoonup u$ (weakly) in $W^{1,2}(\Omega)$ if $u_n \rightharpoonup u$ in L^2 and $D_{u_n} \rightharpoonup D_u$ in $L^2(\Omega)$ for all i .

It follows from (1) and (2) that if $u_n \in W^{1,2}(\Omega)$, $u_n \rightharpoonup u$ in $L^2(\Omega)$ and $\int_{\Omega} |Du_n|^2$ is uniformly bounded by K say, then $u \in W^{1,2}(\Omega)$, $\int_{\Omega} |Du|^2 \leq K$, and $u_n \rightharpoonup u$ in $W^{1,2}(\Omega)$.

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Remark 1.5 (Membership in $W_0^{1,2}(\Omega)$). One can check that if $u \in W^{1,2}(\Omega)$ then $u \in W_0^{1,2}(\Omega)$ iff $\int_{\Omega} D_i u \phi = - \int_{\Omega} u D_i \phi$ for all ϕ in some dense subset of $W^{1,2}(\Omega)$. It follows that membership in $W_0^{1,2}(\Omega)$ is preserved under $W^{1,2}(\Omega)$ weak convergence.

1.2. Dirichlet PDE Problem. The classical Dirichlet PDE problem

$$\Delta u = g \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega,$$

can be reformulated as follows:

Suppose $g \in L^2(\Omega)$. Solve

$$(3) \quad \begin{aligned} &u \in W_0^{1,2}(\Omega), \\ &\int_{\Omega} Du D\phi = - \int_{\Omega} g\phi \quad \forall \phi \in W_0^{1,2}(\Omega). \end{aligned}$$

Using the inner product on $W_0^{1,2}(\Omega)$ and the Riesz representation theorem there exists a unique $u \in W_0^{1,2}(\Omega)$ solving (3). Moreover $u \in W^{2,2}(\Omega)$, and higher regularity follows from higher regularity of g , see [GT83, Ch 8].

The more general classical problem

$$\Delta u = g \text{ in } \Omega, \quad u = f \text{ on } \partial\Omega,$$

is reformulated as follows after setting $w = u - f$:

Suppose $g \in L^2(\Omega)$, $f \in W^{2,2}(\Omega)$. Solve

$$(4) \quad \begin{aligned} &w \in W_0^{1,2}(\Omega), \\ &\int_{\Omega} Dw D\phi = - \int_{\Omega} (g - \Delta f)\phi \quad \forall \phi \in W_0^{1,2}(\Omega). \end{aligned}$$

This is then covered by the previous problem.

One only needs $f \in W^{1,2}(\Omega)$, if one replaces $\int_{\Omega} \Delta f \phi$ by $-\int_{\Omega} Df D\phi(\Omega)$. And similarly, one can also take $g \in W^{-1,2}(\Omega)$, the dual of $W^{1,2}(\Omega)$.

1.3. Dirichlet Energy Minimisation Problem. We solve the energy minimisation problem, examine the minimising sequence, and relate the minimiser to the previous PDE problem.

The classical energy minimisation problem is reformulated as follows.

Suppose $f \in W^{1,2}(\Omega)$. Show there exists $u \in W^{1,2}(\Omega)$ which minimises $\int_{\Omega} |Du|^2$, subject to the constraint $u - f \in W_0^{1,2}(\Omega)$.

(The constraint is a formulation of the condition $u = f$ on $\partial\Omega$.)

Proof and Discussion.

1. Minimising Sequence: Let γ be the inf of $\int_{\Omega} |Du|^2$ such that $u - f \in W_0^{1,2}(\Omega)$. Let $(u_n) \subset W^{1,2}(\Omega)$ be a minimising sequence.

2. L^2 convergent subsequence: Since $W^{1,2}(\Omega) \subset L^2(\Omega)$ is a compact imbedding, there is a subsequence (u_n) such that $u_n \rightarrow u$ (say) in $L^2(\Omega)$ and $\int_{\Omega} |Du|^2 \downarrow \gamma$.

3. $W^{1,2}$ weakly convergent subsequence: From Remark 1.4, $u \in W^{1,2}(\Omega)$, $\int_{\Omega} |Du|^2 \leq \gamma$, and $u_n \rightharpoonup u$ in $W^{1,2}(\Omega)$.

4. u is a minimiser: From Remark 1.5, $u - f \in W_0^{1,2}(\Omega)$ and so $\int_{\Omega} |Du|^2 = \gamma$ since we already know $\int_{\Omega} |Du|^2 \leq \gamma$ from 3.

5. Strong $W^{1,2}$ convergence of the full sequence: We get this directly from 1 by noting

$$\int_{\Omega} |Du_n|^2 + \int_{\Omega} |Du_m|^2 = \frac{1}{2} \int_{\Omega} |Du_n - Du_m|^2 + 2 \int_{\Omega} \left| D \frac{u_n + u_m}{2} \right|^2.$$

The left side terms both decrease to γ , the second right side term is $\geq \gamma$, and so $\int_{\Omega} |Du_n - Du_m|^2 \rightarrow 0$. By Poincare, $\int_{\Omega} |u_n - u_m|^2 \rightarrow 0$.

Then $u - f \in W_0^{1,2}(\Omega)$ since $W_0^{1,2}(\Omega)$ is certainly closed under strong $W^{1,2}$ convergence.

Moreover, $\int_{\Omega} |Du|^2 \leq \gamma$ by Fatou's lemma, and so $\int_{\Omega} |Du|^2 = \gamma$.

6. u satisfies Laplace's equation: Suppose $\phi \in W_0^{1,2}(\Omega)$. Because

$$\int_{\Omega} |D(u + \phi)|^2 = \int_{\Omega} |Du|^2 + 2 \int_{\Omega} Du D\phi + \int_{\Omega} |D\phi|^2,$$

it follows $\int_{\Omega} Du D\phi = 0$. Since otherwise, as small linear terms dominate quadratic terms, we could find ϕ such that $u + \phi$ has less energy than u . $\square$

REFERENCES

- [GT83] David Gilbarg and Neil S. Trudinger, *Elliptic partial differential equations of second order*, 2nd ed., Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], vol. 224, Springer-Verlag, 1983.